CS1231S Assignment #2

AY2023/24 Semester 2

Deadline: Monday, 8 April 2024, 1:00pm

ANSWERS

IMPORTANT: Please read the instructions below.

This is a graded assignment worth 10% of your final grade. There are **six questions** ("question" 0 and five problem-solving questions) with a total score of 40 marks. Please work on it by yourself, not in a group or in collaboration with anybody. Anyone found committing plagiarism (submitting other's work as your own), or sending your answers to others, or other forms of academic dishonesty will be penalised with a straight zero for the assignment, and possibly an F grade for the module. Please see SoC website "Preventing Plagiarism" https://www.comp.nus.edu.sg/cug/plagiarism/.

You are to submit your assignment to **Canvas > Assignments** before the deadline.

Your answers may be typed or handwritten. Make sure that it is legible (for example, don't use very light pencil or ink if it is handwritten, or font size smaller than 11 if it is typed) or marks may be deducted.

You are to submit **a SINGLE pdf file**, where each page is A4 size. Do not submit files in other formats. If you submit multiple files, only the last submitted file will be graded.

Late submission will NOT be accepted. We will set the closing time of the submission folders to slightly later than 1pm to give you a few minutes of grace, but in your mind, you should treat **1pm** as the deadline. If you think you might be too busy on the day of the deadline, please submit earlier. Also, avoid submitting in the last minute; the system may get sluggish due to overload and you will miss the deadline.

Note the following as well:

- Name your pdf file with your **Student Number**. Your student number begins with 'A' (eg: A0234567X). (Do not mix up your student number with your NUSNET-id which begins with 'e'.)
- At the top of the first page of your submission, write your **Name** and **Tutorial Group**.
- To keep the submitted file short, please submit your answers without including the questions.
- As this is an assignment given well ahead of time, we expect you to work on it early. You should submit **polished work**, not answers that are untidy or appear to have been done in a hurry, for example, with scribbling and cancellation all over the places. Marks may be deducted for untidy work.
- It is always good to be clear and leave no gap so that the marker does not need to make guesswork on your answer. When in doubt, the marker would usually not award the mark.
- Do not use any methods that have not been covered in class. When using a theorem or result that has appeared in class (lectures or tutorials), please quote the theorem number/name, the lecture and slide number, or the tutorial number and question number in that tutorial, failing which marks may be deducted. Remember to use numbering and give justification for important steps in your proof, or marks may be deducted.

To combine all pages into a single pdf document for submission, you may find the following scanning apps helpful if you intend to scan your handwritten answers:

* for Android: https://fossbytes.com/best-android-scanner-apps/

* for iphone:
https://www.switchingtomac.com/tutorials/ios-tutorials/the-best-ios-scanner-apps-to-scan-documents-images/

If you need any clarification about this assignment, please do NOT email us or post on telegram, but post on **QnA "Assignments"** topic so that everybody can read the answers to the queries.

Question 0. (Total: 2 marks)
Check that …

- you have submitted a pdf file with your Student Number as the filename. [1 mark]
- you have written <u>both</u> your name and tutorial group number (eg: T02) at the top of the first page of your file. (If you miss either one you will not be given any mark.) [1 mark]

Question 1. Functions (Total: 7 marks)

Define a function $g : \mathbb{Z} \to \mathbb{Z}$ by setting, for each $x \in \mathbb{Z}$,

$$g(x) = \begin{cases} \frac{x}{2}, & \text{if } x \text{ is even;} \\ 5x + 3, & \text{if } x \; is \text{ odd.} \end{cases}$$

(a) Is g injective? Answer "yes" or "no", followed by a proof or counterexample. [2 marks]

(b) Is g is surjective? Answer "yes" or "no", followed by a proof or counterexample. [2 marks]

Given a function $f(x)$, we define the function $f^{(n)}(x)$ to be the result of n applications of f to x, where $n \in \mathbb{Z}^+$. For example, $f^{(3)}(x) = f\Big(f\big(f(x)\big)\Big)$.

(c) What is $g^{(6)}(10)$? Show your working. [1 mark]

We also define the *order* of an input x with respect to f to be the smallest positive integer m such that $f^{(m)}(x) = x$.

(d) What is the order of 2 with respect to the function g? Show your working. [2 marks]

Answers

(a) No, g is not injective. Counterexample: $g(16) = 8 = g(1)$.

(b) Yes, g is surjective.
1. Let $y \in \mathbb{Z}$.
2. Then $2y$ is even (by the definition of even integers).
3. Hence, $g(2y) = \frac{2y}{2} = y$.

(c) $g^{(6)}(10) = g^{(5)}(5) = g^{(4)}(28) = g^{(3)}(14) = g^{(2)}(7) = g(38) = \mathbf{19}$.

(d) The order of 2 is $\mathbf{4}$.

$g^{(4)}(2) = g^{(3)}(1) = g^{(2)}(8) = g(4) = 2.$

Comments from grader (Chin Herng):

(a) Most students who correctly stated that g is not injective is able to give a working counter-example. In attempts to prove injectivity, students first supposed that a, b are integers such that $g(a) = g(b)$, then split into cases depending on a and b are both even or both odd. The issue is that students missed out on the case where one of a and b is odd and the other is even. This is the problematic case that would disprove injectivity.

(b) Students who correctly answered that g is not surjective, but gave an invalid proof, is awarded 1 mark. Invalid proofs mostly arise when students do not work from the first principle: to prove surjectivity, we start from an element in a codomain, then find an element in the domain that maps to it. Some students began by splitting into cases depending on the parity of the input x, did some calculations, but could not arrive at any conclusion.

Some students correctly started from an element y in the codomain. However, some failed to realise that setting $x = 2y$ as the input is already sufficient to show surjectivity, exactly because x is always even. This works regardless of whether y is even or odd! Instead, some students set $x = \frac{y-3}{5}$ in the case where y is odd, which is not only unnecessary but also incorrect. The issue is that $\frac{y-3}{5}$ need not be an integer, let alone an odd integer. Applying g to such an x might not make sense, and even if it does, we might end up dividing x by 2 (because x might be even).

(c) Mostly well done with some minor mistakes in computation. Some students did not take advantage of the given notation, and instead wrote deeply-nested parentheses, bearing the risk that the parentheses messing up, which could cost marks during the final.

(d) Some students attempted to show that $g^{(m)}(2) = g^{(m+4)}(2)$ for all positive integers m. This is not necessary, and from there it is still not immediate by definition that m is the order of 2 with respect to g. Some assumed that $g^{(0)}(2) = 2$, but the entire notation of "n applications of f to x" is only defined for positive integer n only. Lastly, some students worked with f instead of g, but the question never defined f (it only exists as a placeholder function in the given definitions!).

Regardless, most students have the correct intuition and is able to conclude that the answer is 4, but could not produce a sound and simple justification. In this case, it is way more convenient and explicit to exhaust $g(2), g^{(2)}(2), g^{(3)}(2)$ and $g^{(4)}(2)$, then see before your eyes that the first three did not return 2 but the last one did.

Question 2. Functions (Total: 8 marks)

A function $f : \mathbb{R} \to \mathbb{R}$ is said to be *linear* if, and only if,

$$\forall x, y \in \mathbb{R}\, f(x+y) = f(x) + f(y) \quad \text{and} \qquad \forall \lambda, x \in \mathbb{R}\, f(\lambda x) = \lambda f(x).$$

Let $f : \mathbb{R} \to \mathbb{R}$ be a linear function.
Define the relation ~ on $\mathbb{R}$ such that $\forall a, b \in \mathbb{R}\, a \sim b \Leftrightarrow a - b \in f^{-1}(\{0\})$.

(a) Prove that ~ is an equivalence relation. [5 marks]

(b) Provide a well-defined surjective function $\pi : \mathbb{R} \to \mathbb{R}/\sim$. (You need to prove that your function is (1) well-defined, and (2) a surjection.) [3 marks]

Answers

(a)
1. (Reflexivity) (To show: $\forall x \in \mathbb{R}\, x \sim x$.)
   - 1.1. Let $x \in \mathbb{R}$.
   - 1.2. Then $f(x - x) = f(x) + f(-x) = f(x) - f(x)$ (by definition of linear function) $= 0$.
   - 1.3. Hence $x - x \in f^{-1}(\{0\})$, and hence $x \sim x$.
2. (Symmetry) (To show: $\forall x, y \in \mathbb{R}\, (x \sim y \Rightarrow y \sim x)$.)
   - 2.1. Let $x, y \in \mathbb{R}$ such that $x \sim y$.
   - 2.2. Then $f(x - y) = 0$ (by definition of $\sim$)
   - 2.3. Also, $f(y - x) = f(y) - f(x)$ (by definition of linear function)
     $= -\big(f(x) - f(y)\big) = -f(x - y) = 0$.
   - 2.4. Hence, $y \sim x$.
3. (Transitivity) (To show: $\forall x, y, z \in \mathbb{R}\, (x \sim y \wedge y \sim z \Rightarrow x \sim z)$.)
   - 3.1. Let $x, y, z \in \mathbb{R}$ such that $x \sim y$ and $y \sim z$.
   - 3.2. Then $f(x - y) = 0 = f(y - z)$ (by definition of $\sim$)
   - 3.3. Then $f(x - z) = f\big((x - y) + (y - z)\big) = f(x - y) + f(y - z)$ (by definition of linear function)
     $= 0 + 0 = 0$.
   - 3.4. Hence $x \sim z$.
4. Therefore, $\sim$ is an equivalence relation.

(b)
1. Define $\pi : \mathbb{R} \to \mathbb{R}/\sim$ such that $\pi(r) = [r]$ where $r \in \mathbb{R}$ and $[r]$ is the equivalence class of r w.r.t. $\sim$.
2. (To show well-definedness)
   - 2.1. Let $r, s \in \mathbb{R}$. If $r = s$, then $\pi(r) = [r] = [s] = \pi(s)$.
   - 2.2. Hence, π is well-defined.
3. (To show surjectivity)
   - 3.1. Let $A \in \mathbb{R}/\sim$.
   - 3.2. A is non-empty, so take $a \in A$.
   - 3.3. Then $\pi(a) = [a] = A$.
4. Therefore, π is a well-defined surjective function.

Comments from grader (Ikhoon):

FYI: Motivation of this question is from the first isomorphism theorem, which says that under sufficiently good algebraic structure, $\mathbb{R}$ quotient by kernel of the map (namely, inverse image of 0 in this context) is isomorphic to its image (one can prove that there exists a bijection $f : \mathbb{R}/\sim \to f(\mathbb{R})$). In this scenario, the structure is more specific since the function is given to be linear.

(a) Very well attempted. It was an easy question checking whether you know the definition of equivalence relation.

(b) Very poorly attempted. It seems like the vast majority of students are confused with the definition of well-definedness. I want to make a remark that well-defined means $x = y \Rightarrow f(x) = f(y)$, or alternatively, if (x, y_1) and (x, y_2) are in the graph of f, then $y_1 = y_2$.

At the minimum, one needs to say that the image must be unique by property of equivalence classes and with the fact that $x \sim x$, equivalence class partitions the set with none of them being empty.

I was very puzzled that around 90% of the class tried to prove that $x \sim y \Rightarrow \pi(x) = \pi(y)$ (this is just re-stating the definition of equivalence class), which assumes $x = y$ iff $x \sim y$. This is a totally incorrect claim. If f is a zero function, then every value in $\mathbb{R}$ is in the same equivalence class, so $x \sim y$ may not imply $x = y$.

Question 3. Mathematical Induction (Total: 7 marks)

Let $v_0, v_1, v_2, \cdots$ be a sequence defined as follows:

$v_0 = 0$, $\quad v_1 = a$ for some $a \in \mathbb{Z}^+$, and $\quad v_n = v_{n-1} + v_{n-2}$ for $n \geq 2$.

Let $w_n = v_{n-1}^2 - v_{n-2}v_n$ for $n \geq 2$.

(a) Write the values of w_2, w_3, w_4 and w_5. You do not need to show your working. [2 marks]

(b) Make a conjecture for an *explicit formula* (also known as *closed formula*) for w_n in terms of a and n, for $n \geq 2$. [1 mark]

(c) Prove your conjecture in (b) using mathematical induction. (Follow the structure of mathematical induction proofs we used in class. Please do not skip steps or marks will be deducted.) [4 marks]

Answers

(a) $w_2 = a^2, w_3 = -a^2, w_4 = a^2, w_5 = -a^2$.

(b) $w_n = (-1)^n a^2$ for $n \geq 2$.

(c)

1. Let $P(n) \equiv (w_n = (-1)^n a^2)$ for $n \geq 2$.
2. Basis step: $n = 2$
   2.1. $w_2 = v_1^2 - v_0 v_2 = a^2 - 0 \cdot a = a^2 = (-1)^2 a^2$.
   2.2 Hence, $P(2)$ is true.
3. Inductive step: Assume $P(k)$ for some $k \geq 2$, i.e. $w_k = (-1)^k a^2$ for some $k \geq 2$.
   3.1. $w_{k+1} = v_k^2 - v_{k-1}v_{k+1}$ (by the definition of w)
   3.2. $= v_k^2 - v_{k-1}(v_k + v_{k-1})$ (by the definition of v)
   3.3. $= v_k^2 - v_{k-1}v_k - v_{k-1}^2$ (by basic algebra: distributive law)
   3.4. $= v_k^2 - v_{k-1}v_k - (v_{k-1}^2 - v_{k-2}v_k) - v_{k-2}v_k$ (by basic algebra)
   3.5. $= v_k^2 - v_{k-1}v_k - w_k - v_{k-2}v_k$ (by the definition of w)
   3.6. $= v_k^2 - v_{k-1}v_k - (-1)^k a^2 - v_{k-2}v_k$ (by the inductive hypothesis)
   3.7. $= v_k^2 - v_k(v_{k-1} + v_{k-2}) - (-1)^k a^2$ (by basic algebra: distributive law)
   3.8. $= v_k^2 - v_k v_k - (-1)^k a^2$ (by the definition of v)
   3.9. $= -(-1)^k a^2$ (by basic algebra)
   3.10. $= (-1)^{k+1} a^2$ (by basic algebra)
   3.11. Hence $P(k+1)$ is true.
4. Therefore, $P(n)$ is true for $n \geq 2$.

Question 4. Cardinality (Total: 9 marks)

Let A, B and C be sets.

(a) Prove or disprove: $(A \setminus B) \cup (A \cap B) = A$.

For each of the following statements, state whether it is TRUE or FALSE. If true, provide a proof. If false, provide a counterexample. You may use the following propositions without proof:

Proposition 1: If A is finite, then $A \cap B$ is finite.

Proposition 2: If A and B are finite, then $A \cup B$ is finite.

(b) If A is countably infinite and B is finite, then $A \setminus B$ is countably infinite.

(c) If A and B are both countably infinite, then $A \cap B$ is countably infinite.

(d) Suppose $A \subseteq B \subseteq C$. If A and C are both countably infinite, then B is countably infinite.

Answers

(a) [2 marks] Proof:

1. $(A \setminus B) \cup (A \cap B) = \{x : x \in A \wedge x \notin B\} \cup \{x : x \in A \wedge x \in B\}$ (by definitions of set difference and intersection)
2. $= \{x : (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in B)\}$ (by definition of set union)
3. $= \{x : x \in A \wedge (x \notin B \vee x \in B)\}$ (by distributive law)
4. $= \{x : x \in A \wedge true\}$ (by negation law)
5. $= \{x : x \in A\}$ (by identity law)
6. $= A$

(b) [3 marks] **True.**

1. Note that $A \setminus B \subseteq A$ (by specialisation on the definition of set difference:
$$A \setminus B = \{x \in U : x \in A \wedge x \notin B\})$$
2. $A \setminus B$ is countable as $A \setminus B \subseteq A$ (by Theorem 7.4.3).
3. $A \setminus B$ cannot be finite (see proof below).
   - 3.1. Suppose $A \setminus B$ is finite.
   - 3.2. $A \cap B$ is finite since B is finite (by proposition 1).
   - 3.3. Then $(A \setminus B) \cup (A \cap B)$ is finite (by proposition 2).
   - 3.4. But $(A \setminus B) \cup (A \cap B) = A$ (by part (a)), which contradicts that A is infinite (given).
   - 3.5. Hence $A \setminus B$ is infinite.
4. Since $A \setminus B$ is countable (from 2) and infinite (from 3.5), therefore $A \setminus B$ is countably infinite.

Theorem 7.4.3: Any subset of a countable set is countable.

(c) [2 marks] **False.** $\mathbb{Z}^-$ and $\mathbb{Z}^+$ are countably infinite sets, but $\mathbb{Z}^- \cap \mathbb{Z}^+ = \emptyset$ is a finite set.

Theorem Cardinality.1: Any subset of a finite set is finite.
Theorem 7.4.3: Any subset of a countable set is countable.

(d) [2 marks] **True.**

1. A is countably infinite (given), so A is infinite, and hence B is infinite (by contrapositive of Theorem Cardinality.1)
2. C is countably infinite (given), so C is countable, and hence B is countable (by Theorem 7.4.3.)
3. Therefore, B is countably infinite.

Comments from grader (Udhaya):

Question was generally well done, students shown understanding of the theorems provided with a small group resorting to more tedious functions and sequence argument.

(a) Well done. Justifications were well provided. Uncommon mistakes include some students writing universal set U as TRUE possibly due to mix up with Theorem 2.1.1. Additionally, some students opted for an element argument to prove this, which can be tedious as it requires proving both sides by the definition of set equality.

(b) Common mistakes stem from a misunderstanding between countable and countably infinite.

  (i) A common error observed was the misinterpretation of what it means for a set to be countably infinite. It is important to understand that a set can fall into one of three categories: **finite, countably infinite, or uncountable.** Therefore, to demonstrate that a set is countably infinite, it is crucial to establish that it is neither finite nor uncountable. Several responses mistakenly classified sets as countably infinite simply because they were countable, neglecting to recognize that countable implies either finite or countably infinite. Remember, according to the definition, a set is countable if and only if it is finite or countably infinite.

  (ii) The use of the sequence argument was common among students, and while it is a valid approach to demonstrate that a set is countable, it doesn't necessarily prove that the set is countably infinite which many students thought. Proposition 9.2 states the use of the sequence argument to establish that a set is **countable**. However, to fully conclude that the set is countably infinite, it is necessary to show it is also infinite which is commonly missed out.

(c) Well done. Any valid counter example suffices. A mistake includes leaving the marker to interpret set sequences as even or odd i.e. student must explain set {2,4,6,8, ..} is even by using proper set notation and even number definition instead of just writing out the sequence.

(d) Common mistakes are similar to (b) as well.

  (i) Some missed out on justifications by not citing relevant theorems, up to -1 mark was deducted for this (-0.5 for each justification for infinite and countable).

  (ii) Some students stopped at showing that B is countable and hence it must be countably infinite. As explained in remark for part (b), they must continue to show that B is infinite.

Question 5. Counting and Probability (Total: 7 marks)

(a) How many integer solutions are there for the following equation, where each $x_i \geq 1$?

$$x_1 + x_2 + x_3 = 10.$$

Show your working instead of just writing out the final answer. [1 mark]

(b) The figure on the right shows a combination lock with 40 positions. To open the lock, you rotate to a number in a clockwise direction, then to a second number in the counterclockwise direction, and finally to a third number in the clockwise direction.

If consecutive numbers in the combination cannot be the same, how many three-number codes can there be? Write your answer with explanation. [1 mark]

(c) A recurrence relation is an equation that defines a sequence based on a rule the gives the next term as a function of the previous term(s). For example, the factorial function can be defined by this recurrence relation:

$$0! = 1 \quad \text{and} \quad n! = n \times (n-1)! \text{ for } n > 0.$$

In an orientation game with girls and boys, chairs are placed in a row and pupils are to sit on them, one person on each chair. The rule, however, states that no two boys are allowed to sit next to each other.

Let there be $n \geq 1$ chairs, and let $f(n)$ be the number of ways the chairs can be filled. For example, let G denote a girl, and B a boy, then for $n = 2$, the chairs may be filled with GG, GB or BG, a total of 3 ways, hence $f(2) = 3$. (i) Write a recurrence relation for $f(n)$, including its initial condition(s), and (ii) what is $f(12)$? [2 marks]

(You do not need to explain how you obtain (i), and do not need to show the working for (ii).)

(d) A bowl contains three coins, two of which are normal coins and one is a two-headed coin. You pick one coin at random and toss it. What is the probability that you get a head? Show your working and write your answer as a fraction. [2 marks]

(e) A bowl contains three coins, two of which are normal coins and one is a two-headed coin. You pick one coin at random, toss it and get a head. What is the probability that the coin is two-headed? Show your working and write your answer as a fraction. [1 mark]

Answers

(a) $y_1 + y_2 + y_3 = 7$ where each $y_i \geq 0$. Hence $n = 3, r = 7$, and $\binom{r+n-1}{r} = \binom{9}{7} = \mathbf{36}$.

(b) $40 \times 39 \times 39 = \mathbf{60840}$. The second number cannot be the same as the first, hence 39 ways. The third number cannot be the same as the second, hence 39 ways.

(c) (i) $f(1) = 2, f(2) = 3, f(n) = f(n-1) + f(n-2)$ for $n > 2$.
(ii) $f(12) = 377$. (Sequence is 2,3,5,8,13,21,34,55,89,144,233,377, ...)

(d) Let N be the event that you pick a normal coin and A be the even that you pick the abnormal (two-headed) coin. Then $P(H|N) = 0.5$ and $P(H|A) = 1$.
Therefore, $P(H) = P(H|N) \cdot P(N) + P(H|A) \cdot P(A) = \frac{1}{2} \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \mathbf{\frac{2}{3}}$

(e) $P(A|H) = \frac{P(H|A) \cdot P(A)}{P(H)} = \frac{1 \cdot \frac{1}{3}}{\frac{2}{3}} = \mathbf{\frac{1}{2}}$

Comments from grader (Justin):

(a) Well attempted.

(b) Generally well attempted. Some mistakes are:
- (i) Assumed that the third number could not be the same as the first number and hence $40 \times 39 \times 38$.
- (ii) Careless mistake, putting 60480 instead of 60840.
- (iii) No working/explanation even though question has explicitly specified to do so.

(c) Many careless mistakes:
- (i) Some did not give the base/initial cases, in which no marks were awarded since the base/initial cases are crucial when defining the recurrence.
- (ii) Some also had wrong base/initial cases ($n = 0$ and $n = 1$), which was also marked wrong as the question had also explicitly stated that $n \geq 1$.
- (iii) Many could state the correct base cases, however, did not specify that $f(n) = f(n-1) + f(n-2)$ is for $n > 2$, in which half mark was deducted.
- (iv) Inconsistent use of x and n.

(d) Generally well attempted.

(e) Generally well attempted.

Graders
Q0: 2 marks.
Q1 (Functions): 7 marks. Graded by Chin Herng. chinherng@u.nus.edu
Q2 (Functions): 8 marks. Graded by Ikhoon. e0908299@u.nus.edu
Q3 (Mathematical Induction): 7 marks. Graded by Vladimir. e0974181@u.nus.edu
Q4 (Cardinality): 9 marks. Graded by Udhaya. e0958337@u.nus.edu
Q5 (Counting and Probability): 7 marks. Graded by Justin. justintanwk@u.nus.edu
Total: 40 marks

=== End of paper ===